PRODUCT REQUIREMENTS DOCUMENT

Field Sales Tracker

Verified field activity, client visits and meeting documentation for Carbon Car Care's outdoor B2B sales team

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Markets	Nigeria, Zambia, Malawi
Platform	Standalone offline-capable PWA; Lovable + Supabase
Status	Revised draft for stakeholder sign-off

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In one line

A standalone, offline-capable field-sales app for attendance, physical client visits, meeting documentation, client/branch registration, exceptions and audited management oversight.

Scope correction - v1.2 MVP update

Follow-up task management, quotations, opportunities, orders, sales conversion, pipeline values and revenue analytics are deferred to a future separately approved phase. Simple meeting purpose, outcome and free-text next step remain in the MVP.

1. Overview and product principles

Problem. Carbon Car Care's outdoor B2B sales team lacks consistent, auditable records of attendance, site presence, meetings and approved exceptions across Nigeria, Zambia and Malawi.

Objective. Provide location- and photo-supported field activity evidence, reliable offline operation, documented meeting outcomes, exception reviews and clear management visibility. GPS alone does not prove a client meeting.

1.1 Scope and agreed decisions

Item	Confirmed v1.2 MVP decision
Markets	Nigeria, Zambia and Malawi; local work calendars and IANA time zones.
Customers	B2B fleet businesses and wholesale buyers; organisations may have multiple branches.
Devices	Company-issued phones; approved inventory, assignment and replacement controls.
Platform	Mobile-first installable PWA using Lovable and Supabase.
Connectivity	Offline-first capture with queued records and controlled resynchronisation.
Integration	Standalone MVP; versioned CSV export only. ERP/CRM connection is future work.
Visit planning	Salespeople select client visits; no assigned routes or continuous tracking.
Sales features	Meeting purpose, discussion, outcome and a free-text next step only in MVP.
Verification	Geofence/photo/time/log rules; separate selective client audits and review of exceptions.

1.2 Product principles

- **Evidence, not absolute proof:** location, photographs and timestamps are supporting signals subject to verification.
- **Clear states:** attendance, location verification, meeting audit and exceptional approvals are independent.
- **Immutable originals:** correction requests and reviewer decisions never replace captured evidence.
- **Fair exceptions:** approved office work, leave, reception failures and device breakdowns have explicit procedures.
- **Country segregation:** country managers see their own market; authorised HQ sees all markets.

1.3 Outside the MVP

Quotation creation/status/value, follow-up tasks and reminders, orders, opportunities, funnel/conversion analytics, sales pipeline/revenue reporting, invoicing, payment collection, fuel/allowances, continuous tracking, biometric face matching, route optimisation, client portals and live ERP/CRM integrations.

> **Counting policy**
>
> A site visit may be Location Verified without client confirmation if required evidence passes. An approved GPS/device failure is logged separately as Manager Approved - Location Unverified.

2. Users, roles and governance

Three operational roles are included. Data access is restricted by assigned user and market, with authorised cross-market HQ oversight and append-only approval histories.

Role	Scope	Allowed MVP actions
Salesperson	Own records and assigned market	Start/End Day, visit and meeting logs, register branches, request corrections and GPS/device exceptions, propose non-visit workdays.
Country Sales Head	One market	Review branch/pin approvals, attendance exceptions, disputed visits, audit calls and non-visit workdays; approve device reassignment; view field dashboards.
HQ Admin	All markets	Manage staff, work calendars, devices, rule thresholds, exports, retention and country-level permissions; review escalations.

2.1 Permissions and audit policy

- Only the **salesperson** may create and submit their own normal visit, missed-visit or GPS/device-failure exception log, including after device access is restored.
- A Sales Head may **approve, reject or escalate** submitted exceptions; managers cannot create or rewrite salesperson visit logs.
- Salespeople cannot silently edit or delete submitted evidence. Corrections are linked requests, preserving original data.
- Approval decisions record actor, reason, date, evidence, previous status and any escalation. Self-approval is not permitted.
- HQ can archive/deactivate under retention policy; routine evidence deletion is unavailable to operational users.

2.2 Work calendars and approved non-visit days

Area	MVP requirement
Work calendars	Market-specific working hours, weekends, public holidays and approved absence/leave.
Non-visit activities	Salesperson requests office assignment, proposal preparation, training, virtual meeting, management meeting or other, with reason and optional evidence.
Reporting rule	Manager-approved non-visit time counts as attendance but never as a GPS-verified client visit.
Manager oversight	Review decisions and pending requests displayed in exception/attendance dashboards.

2.3 Company-issued devices

HQ records asset tag, make/model, OS, assigned salesperson, issue/return dates and revoked state. Replacement requires approval. A PWA session token is not tamper-resistant device attestation; assess company device management separately.

3. Salesperson daily flow

Primary flow: Start Day, select branch, Check In, capture premises photo, Check Out, document meeting, repeat, End Day. Offline and exception paths are visible throughout.

Step	Action and system behaviour
1. Start Day	Capture GPS and live front-camera selfie; identify workday type and authorised company device. Offline records remain provisional until synced.
2. Select branch	Search organisations and branches or Nearby (default 2 km where pinned). Show pending visit logs, sync problems and exceptions; **no follow-up task list**.
3. Check In	Take fresh GPS reading, apply branch geofence, capture mandatory live premises photo; retry or request exception for unreliable readings.
4. Meeting	Record evidence and visit elapsed duration. No always-on location tracking.
5. Check Out	Capture fresh GPS and duration. Outside-geofence check-out is permitted but flagged for review.
6. Meeting Log	Record person met, purpose, products discussed, outcome, notes and a free-text next step. Aim to submit within 2 hours; late records are flagged. No quote/order values or follow-up tasks.
7. End Day	Capture GPS where possible; preserve auto-close and reconciliation flags at configured country day boundary.

3.1 Visit state progression

Draft / Capturing -> **Pending Sync** (if offline) -> **Pending Verification** -> **Location Verified / Location Unverified / Pending Client Approval**. Separately track meeting-log completeness, audit results, disputes and approved exceptions.

3.2 Visit counting policy

- Only visits with eligible branch, accepted location reading, premises photo, check-in/out timestamps and completed meeting log count toward standard verified visits.
- Approved GPS/device failures are classified **Manager Approved - Location Unverified** and reported separately.
- Client audit denial creates **Disputed - Pending Review**, not an automatic violation.
- Pending clients, unsynced records and insufficient evidence are provisional until processed.

3.3 Failure and correction path

If GPS is inaccurate, offer fresh readings and a salesperson-created exception submission. If the phone fails, the salesperson submits the missed-visit log after access is restored, with claimed actual visit time, delayed-submission explanation and any evidence. Managers only approve/reject; they never create visit logs.

4. Client and branch registration

A client organisation can own multiple branches. The business master and visit location must therefore be separate entities. Phone numbers may legitimately be shared by headquarters and branches.

4.1 Master import before go-live

- Country Sales Heads supply a market CSV with parent company, branch name, client type, address/landmark, contacts, contact phone and optional GPS, expected fleet size or monthly buying volume.
- HQ validates required columns, normalises phone numbers, detects duplicates and previews import errors before confirmation. Every import receives an immutable batch ID and row-level results.
- Existing branches with a verified coordinate and manager sign-off may become Active; otherwise use Location Pending. Initial coordinates should not automatically be treated as independently verified.
- At first branch visit, a GPS point with accuracy target 50 m and live premises photo is proposed, then the Sales Head approves or adjusts it with a reason.

4.2 Registration on the ground

- After Start Day, salesperson chooses existing parent company or creates a proposed new company, then enters branch name, address, type, contact, size/volume and a live premises photo.
- Capture GPS with target accuracy 50 m; if unavailable, allow pending-location registration but prohibit location-verified visit status until the pin is approved.
- Check exact branch/address, similar name and nearby clients within 100 m; a shared phone is a warning, not a hard rejection. Same parent/phone across distinct branches is supported.
- Create Pending Approval and allow the visit to proceed as pending evidence. Sales Head approves/rejects within two working days; rejection leaves visit in audit history as unverified.

4.3 Client/branch and pin controls

Entity or rule	Requirement
Parent company	Company ID, legal/trade name, industry/type, assigned market, main contacts; may have multiple branches.
Branch	Branch ID, parent ID, distinct location and geofence, address, photos, approval state, market and contact links.
Contacts	Branch and/or company level; one phone may be associated with more than one branch. Flag duplicates with contextual match evidence.
Pin change	Save original and revised coordinates, actor, reason, effective time and affected visits; never silently re-score prior historical evidence using a new pin.
State	Location Pending / Pending Approval / Active / Rejected / Inactive. Separate record approval from geographic verification.
Audit	Sales Head reviews branch/map/photo and possible duplicate warnings; HQ sees pending backlogs and SLA breaches.

5. Offline records and synchronisation

The PWA must support loss of connectivity, but a browser cannot guarantee that local data survives user deletion, OS cleanup, app reinstall or device loss. The system should be built for resilient recovery and transparent uncertainty, not promises of tamper-proof offline storage.

5.1 Offline flow

- Cache the app shell and the minimum assigned-market clients, branches, contacts and rule versions for approved company devices. Cache revisions include `downloaded_at` and validity version.
- Persist each action in IndexedDB with a generated UUID, device session ID, ordered sequence, relevant event IDs, photo references, GPS and reported accuracy, local time and timezone, and previously obtained server-time anchor where available.
- Queue write must complete locally before the app displays Captured Offline. Capture and media metadata are committed together, or the user sees a failed-save warning.
- Sync on reconnect/app open/manual Sync now, plus browser background sync where supported. Upload dependencies in order but permit idempotent retry and per-item error handling.
- Server receipt time is authoritative for receipt, not necessarily for the original offline event time. Offline capture time remains a device claim, assessed against sequence, server anchors and plausibility.
- On sync, server checks identity, event order, market, geofence, duplicate IDs, photos, device/session status and unusual travel/clock signals; returns an explicit result for each event.

5.2 Offline integrity and conflict resolution

Scenario	Required action
Retry / duplicate upload	Event UUID idempotency key: first accepted write retained; retries return the same result without duplicating activity.
App/browser restart	Reopen persisted queue; resume retries. In-app elapsed timer is not presumed accurate across restarts.
Phone clock changes	Record claimed local time and last server-time anchor, flag material mismatch; do not represent elapsed browser time as a secure clock.
Client pin changed	Validate against the approved pin version captured for the visit; if client was provisional, require manager review.
Offline new branch duplicate	Re-run uniqueness/conflict checks on server; quarantine ambiguous records rather than silently merging.
Media upload fails	Retain event as Pending media / Unverified until photo uploaded and validated; allow retry with progress indicator.
Browser storage cleared	Missing unsynced data cannot be recovered from server. Flag device reset and require manager reconciliation; no fabricated replacement GPS.
Missed day-end sync	Show sync-late exception with last known device contact. Distinguish delay due to reception from deliberate withholding.

5.3 Offline risk rules

Default same-day sync target is 23:59 market local time, with exceptions reviewed where network availability is poor. More than 30% offline visits per week triggers prioritised audit, not automatic misconduct. The server can show only **last received sync** and **last reported pending count**; it cannot know live queue size on a disconnected phone.

> **Critical design constraint**
> Local queue locks and logout blocks are user-experience measures, not security guarantees. Device management, server checks, reconciliation and client audits are the actual risk controls.

6. Verification, flags and audit controls

Separate hard blocks (provable invalid application actions), soft evidence classifications (uncertain location) and risk flags (investigate without presuming dishonesty). All thresholds are configurable per market and logged when changed.

6.1 Hard blocks and exception routes

Control	Rule and behaviour
Active workday	Normal visits require an open attendance day. A late visit log after device failure is submitted by the salesperson as a separate reviewable exception.
Device/user identity	Approved account, market, session and company-issued asset. Device token is an identifier, not tamper-proof proof of the physical device.
One open visit	Disallow parallel normal visits. Salesperson submits a correction for an abandoned visit; manager may approve or reject, not create a log.
Client pin	Known active branch geofence must be evaluated on fresh GPS. Clearly outside -> retry or location exception, not verified check-in.
Location reading	Default desired GPS accuracy 100 m or better for visits, 50 m for pin proposals. Borderline/bad fixes -> retry or request manager review.
Photo origin	Capture via `getUserMedia` UI; gallery upload disabled for normal user flow. This cannot prove physical originality against sophisticated spoofing.
Record mutation	Client actions are append-only after submission. Manager annotations/corrections are separately logged and controlled.
Checkout location	Always permit checkout; outside branch geofence creates a risk flag rather than trapping the visit.

6.2 Location evidence classification

Recommended default: calculate haversine distance `d` to the branch pin and retain reported horizontal accuracy `a`. A reading with `d + a <= radius` is higher-confidence inside; where `d <= radius` but accuracy overlaps the boundary, classify as uncertain and offer another reading or manager review. Reject verified status for clear-outside readings. Reported GPS accuracy is not an anti-spoofing guarantee.

6.3 Automated flags - attendance and field patterns

Flag	Default trigger / notes
Short visit	Duration under 5 minutes; risk indicator, not automatic fraud.
Checkout outside	Check-out distance beyond approved branch radius; visit can close.
Impossible travel	Implied speed above 80 km/h between captured points; account for clock/sync uncertainty.
Long idle gap	More than 150 minutes between visits during an assigned field-work block; excludes approved non-visit activities.
Late start / early finish	After 08:30 / before 16:30 where scheduled; apply market calendar, leave and exemptions.
No visits / low visits	None by 12:00 or below configurable target; do not flag approved non-field days.
Late meeting log	Submitted more than 2 hours after check-out.
Near attendance origin	Branch within 300 m of Start Day point; review context, no presumption of fraud.

6.4 Automated flags - evidence and data risk

Flag	Default trigger / notes
Duplicate / similar photo	Perceptual image-hash match or high similarity to earlier premises evidence.
Clustered registrations	Same salesperson proposes 3 or more branches within 100 m; legitimate complexes may qualify.
Possible duplicate branch	Similar name/address/nearby branch; shared company phone alone does not prohibit creation.
Repeated same branch	Three or more visits in seven days; manager considers account needs.
Client rejected	Proposed branch rejected; related visit remains unverified.
Late sync / clock anomaly	Not received before same-day cutoff or device time inconsistent with anchors/event sequence.
High offline share	More than 30% of visits weekly offline; risk-based audit priority.
Unapproved device / reset	New device, cleared app state or reassignment pending approval.
Day not closed	Auto-closed at configured day boundary, reconciled if later offline events arrive.
Client meeting disputed	Client audit denies meeting; high-priority review with reply rights.
Missing evidence	Mandatory media/log unavailable; do not count as location verified.
Rejected exception	Manager declines GPS, device failure or non-field-day explanation.

6.5 Audit calls and decision fairness

Select a random 5% of location-verified visits weekly per market; add targeted checks for disputed visits, rejected new clients, high offline share and other unusual patterns. Random and targeted samples must be labelled separately so the audit rate remains interpretable.

Audit outcome	Effect on verification / reporting
Confirmed	Mark meeting Client Confirmed. Record who confirmed, date and contact; location evidence retains separate status.
Not confirmed	Set Disputed - Pending Review. Salesperson may submit explanation; manager investigates photos, contacts, visit facts and timing before final outcome.
Unreachable / unclear	Record inconclusive, schedule appropriate retry and keep current evidence classification; do not automatically label fraudulent.
Confirmed false claim	Record a Confirmed Violation with review note and evidence. Remove the visit from legitimate-meeting totals; leave source events for audit.
Dispute resolved in favour	Mark Resolved - Valid with reviewer justification; preserve original denial and investigation history.

6.6 Review controls

- Each flag enters an exception queue with severity, owner, due date and action history. Outcomes: Open, Under Review, Cleared, Confirmed Violation, or Escalated.
- Review due targets: critical dispute within one working day; ordinary flags within three working days; new branch/pin approval within two working days. Targets configurable by market.
- Confirmed violations require evidence, recorded reviewer and policy reference. Consequences are determined by Carbon Car Care HR policy, not automatically by the software.
- Where face matching is considered later, require a separate privacy and accuracy assessment, appeal process and stakeholder approval. It is not part of v1.2 MVP.

7. Manager dashboards and field reporting

MVP reporting separates attendance, physical visits, evidence quality and manager-reviewed exceptions. It provides operational oversight only; it does not track quotations, orders, pipelines or revenue.

View	Core MVP contents
Live / daily	Scheduled members, attendance/leave/non-visit states, last server sync, open visits, visited branches, evidence status, missing meeting logs, pin/branch approvals and urgent flags.
Weekly	Per salesperson: days worked, approved non-visit days, location-verified visits, unique branches, exception-approved activities, offline percentage, flag closure and audit outcomes.
Quarterly	Unique company/branch coverage, physical visit trends, verified visit rate, exception trends, team attendance and frequency of repeat visits; no commercial funnel.
Salesperson day view	Map/timeline of captured action points, visit duration, photos, accuracy, last sync, manager decisions and outstanding exceptions.
Company / branch history	Approved pins, contact records, visit dates, evidence, meeting summaries and audit/correction history. No quotations or orders.
Exception workbench	Filter by market, rep, severity, unresolved GPS/device events, client disputes, overdue approvals and audit call results.

7.1 MVP KPI definitions

Metric	Calculation / eligibility
Location-verified visits	Distinct completed branch visits passing server location, photo, timing and meeting-log checks; exclude pending/rejected records.
Approved unverified activity	Approved device/GPS exceptions and approved non-visit days displayed in separate categories, never added to verified count.
Meeting audit confirmation	Client-confirmed calls divided by conclusive audit attempts; show unreachable/inconclusive counts separately.
Client/branch coverage	Unique verified visited branch IDs / eligible active branches in selected market/period.
Evidence and sync health	Share of completed visits verified; offline share; late sync, GPS failure and open flag counts.
Review turnaround	Percentage of exception/branch approvals resolved within configurable local SLA.

7.2 Scheduled reports and export

Daily 19:00 local provisional exception email; next-morning reconciliation after pending sync. Weekly Monday HQ export by market, visit verification, attendance and exceptions. CSV includes stable IDs, source, timezone and export schema version, and uses safe formula escaping.

> **Explicit MVP exclusion**
>
> No quote count/value, order count/value, opportunity stages, overdue follow-up tasks, quote-to-order conversion, pipeline or revenue KPIs. These require future approval and a separate data model.

7. Meeting records and future sales modules

7.3 Salesperson meeting log - MVP

Field	MVP rule
Contact person	Select existing contact or add name, role and phone. Required.
Visit purpose	New pitch / Client follow-up meeting / Product discussion / Negotiation / Relationship / Other. Required; descriptive only.
Products discussed	Tyres / Batteries / Lubricants / Fleet servicing / Telematics / Other. Required multi-select.
Meeting outcome	Interest expressed / Information requested / Met decision-maker / No decision-maker / Not interested / Other. Required as a visit note, not a deal status.
Fleet size or volume	Collect during first discovery meeting if available; editable with audit trail.
Next step	Required free text describing the expected next action. No due-date task, reminder, status or owner workflow.
Premises photo	Mandatory live camera photograph for normal location-verified visits.
Business card photo	Optional live-camera capture, subject to privacy rules.
Notes	Optional free text. Discourage collecting unnecessary sensitive personal data.

7.4 Deferred commercial modules - not part of MVP

Future module	What may be specified later
Follow-up management	Task ownership, due dates, reminders, overdue queues, completion results and follow-up reporting.
Quotation tracking	Requests, issued quotes, document references, monetary values, currencies and quote statuses.
Order reporting	Salesperson-reported orders, manager verification, external order references and commercial values.
Opportunity / pipeline	Opportunity stages, quote-to-order conversion, commercial funnels and sales performance dashboards.
ERP/CRM integration	Optional future API sync and reconciliation after a separate specification and approval.

> **MVP boundary**
>
> MVP may record that a customer expressed interest or requested information as meeting narrative. It must not create commercial workflow records, monetary sales metrics, reminders or automatic follow-up tasks.

7.5 Conditions for a later sales phase

- Require separate scope approval, ownership, acceptance tests, data migration design and delivery estimate.
- Evaluate whether the existing ERP/CRM should be the system of record before building overlapping quotation or order features.
- Keep launch operational reports free of commercial KPIs until a future module is deployed and tested.

8. MVP data model and event states

All launch tables use stable UUID identifiers, country/market segregation, traceable creation times and append-only evidence events. Commercial workflow entities are excluded from the MVP schema.

8.1 Core MVP entities

Table	Purpose and essential columns
markets	id, name, IANA_timezone, currency, rule_version.
users	id, role, market_id, employee_id, name, contact, active; country-scoped permissions.
devices	id, asset_tag, assigned_user, model, OS, session token, issue/revoke/approval state.
work_schedules	id, market/team/user, working date and hours, holiday, authorised leave/activity.
attendance_days	id, user_id, day, start/end event IDs, workday type, attendance state and sync status.
organisations	id, market_id, legal/trade name, category, status and optional owner.
branches	id, organisation_id, name, address, lat/lng, geofence radius, pin version, approval state.
client_contacts	id, organisation_id, nullable branch_id, name, role, phone/email. Phone can be shared.
client_changes	id, organisation/branch, old/new value, changed_by, approved_by, reason, effective time.
visits	id, user, branch, attendance ID, check-in/out event IDs, verification and review statuses.
visit_photos	id, event/visit ID, photo kind, private storage key, hash and stamped-copy pointer.
meeting_logs	id, visit, contact, purpose, product list, narrative outcome, next-step text, notes, submit time.
activity_events	id, device, user, action type, location+accuracy, client time, server receipt, offline flag, UUID, sequence, rule version.

8.2 Controls and operational support entities

Table	Purpose and essential columns
exceptions	id, subject, creator salesperson ID, stated reason, evidence, submitted/review status, reviewer and decision.
flags	id, subject, rule code/values, severity, status, reviewer and rule version.
audit_checks	id, visit, random/risk category, contact, result, review note and dispute link.
approval_audit	id, decision actor, subject, before/after state, time, reason and evidence.
import_batches	id, source checksum, operator, market, validation and row outcomes.
export_jobs	id, export type, market, schema version, operator, completion and file reference.

> **No MVP commercial tables**
>
> Do not create follow_up_tasks, opportunities, quotations or reported_orders in the MVP database or Lovable generation prompts. Define them later only if the future sales phase is approved.

8. Data integrity and status rules (continued)

8.3 Evidence and referential integrity

- Store each captured action as an immutable event with client-generated UUID and server receipt timestamp. Server validations and manager decisions drive read models and KPIs.
- Use PostgreSQL foreign keys and unique event IDs; enforce idempotency and ordered dependency resolution for offline uploads.
- Apply `market_id` country scoping through Postgres RLS and Edge Functions, including private media and export access.
- Retain photos privately and create separate server-stamped copies for controlled exports. Watermarks alone do not establish provenance.
- Version branch pins and verification rules so later edits do not silently reclassify historical visits.
- Any manager approval or correction appends a dated decision and never replaces the salesperson's original event.

8.4 Canonical MVP status definitions

Record type	States and reporting behaviour
Visit evidence	Capturing / Pending Sync / Pending Review / Location Verified / Location Unverified / Pending Client / Rejected. Only Location Verified counts as standard verified visits.
Meeting log	Missing / Submitted on time / Submitted late / Needs Clarification; capture narrative meeting outcome only.
Client audit	Not Audited / Confirmed / Disputed / Resolved Valid / Confirmed False / Inconclusive, separately from GPS status.
GPS/device exception	Draft / Submitted / Under Review / Approved / Rejected / Escalated. Submitted by salesperson only; approved status remains Location Unverified.
Attendance	Scheduled / Started / Ended / Auto-Closed / Approved Non-Visit / Approved Leave / Absent / Pending Sync.
Organisation/branch	Location Pending / Pending Approval / Active / Rejected / Inactive, with pin and branch history.

8.5 Data model intentionally excluded

No follow-up task, quotation, order, sales opportunity, deal-stage or conversion entities are required in MVP. The meeting log's free-text next step is informational only; it creates no reminder, assigned task or due-date service.

> **Data contract for future work**
>
> Preserve organisation, branch, contact, meeting-log and visit UUIDs, field meanings and CSV versions so later approved sales modules may reference these records without modifying original field evidence.

9. Technical, security and non-functional requirements

Architecture remains Lovable frontend, Supabase Auth/Postgres/Storage/Edge Functions and installable PWA. Sensitive validation happens on the server. Company-issued devices allow a controlled test matrix and optional device-management policy.

9.1 Architecture and technical controls

Layer	Requirement
Client / PWA	Mobile-first, high contrast, large touch targets; service worker app shell; IndexedDB offline queue with clear sync states. Works without maps when offline.
Location	Geolocation API with `enableHighAccuracy`; request fresh readings and store lat/lng, horizontal accuracy, requested/captured timestamps and pin version.
Camera	`getUserMedia` for selfie/premises; no regular user file picker. Explain device/browser permission failures and enable evidence exception workflow.
Backend	Supabase Edge Functions validate every event, role, client state, idempotency, rule versions, geo conditions and exception eligibility.
Identity	Supabase authentication; company-managed credentials, secure session expiration, reauthentication for sensitive actions and server-side device allow-list.
Authorisation	RLS at row and storage access levels; all critical writes use server-side checks to prevent role bypass. Security tests must cover direct API calls.
Data / media	Postgres foreign keys and audit tables; private Supabase Storage buckets, MIME/size validation, image similarity signals and stamped export copies.
Maps	Leaflet with approved OSM tile service or provider meeting use limits; manager map online, offline client list available without tile dependence.
Reports	Scheduled Edge Function for 19:00 local digest; next-day reconciliation; CSV with stable schema and safe handling of formulas in export cells.
Operations	Monitoring for failed jobs, upload errors, excessive flag volume, queue conflicts, authentication anomalies, storage growth and export failures.

9.2 Data protection and employee transparency

- Publish a company-device and employee-monitoring notice: location at explicit actions only, purpose, access roles, audit approach, retention and complaint/contact channel.
- Assess applicable Nigerian, Zambian and Malawian personal-data and cross-border transfer obligations with company legal/compliance before launch. Do not claim legal compliance from software architecture alone.
- Set approved retention durations for raw selfies, premises/business-card photos, GPS events, audit records and exports before go-live. Include secure deletion/export procedures and documented backups.
- Restrict full-resolution photos, personal phone numbers and home-origin points to authorised roles; mask contact fields in bulk reporting where possible.
- No face matching in MVP. If later introduced, undertake separate biometric impact assessment and formal approval.

9. Non-functional requirements (continued)

9.3 Production acceptance targets - proposed defaults

Quality area	Proposed acceptance measure
Basic performance	On approved pilot devices, app home screen responds within 3 seconds on a usable 4G link after first load; offline cached shell launches within 5 seconds.
Capture reliability	Offline event + associated media save is acknowledged only after local transaction succeeds; failure shows retry and preserves previous records.
Sync correctness	Retried event UUID never creates duplicates; dependent events resolve in order; each event returns Accepted / Needs Review / Rejected with reason.
Offline persistence	Survives a normal app/browser restart in QA; document unavoidable loss risk from browser data removal, device reset and storage exhaustion.
Security	Zero critical/high findings from agreed auth/RLS/Storage test suite at pilot gate; all privileged decisions logged.
Location UX	Clear prompt for denied permission, poor accuracy, inability to get GPS fix and manager exception; no false location verification.
Data reporting	Daily KPI reconciliation re-runs after delayed sync, with auditable changes; provisional vs final reporting marked.
Accessibility	Outdoor readable, touch targets sized for mobile use; no essential action depends only on colour.
Recovery	Backups, restoration rehearsal, alerting contacts and runbook defined by IT/operations before rollout.

9.4 Device/browser compatibility gate

Confirm selected company handset model(s), Android/iOS version and management policy before committing to a pure PWA. Test GPS in urban canyons and business parks, browser resume/kill, camera permission switching, poor network, storage quotas, airplane mode, 2G/3G, clock changes, power saving and device reassignment. Native wrapper remains a **conditional future option**, not a guaranteed solution to all GPS spoofing.

> **Delivery risk**
> A secure, fault-tolerant offline application is the primary engineering risk. Test it early with real phones before committing to a Lovable-only timeline. The original 4-5 week estimate is a hypothesis, not an approved production schedule.

10. MVP scope, rollout and success measures

Launch scope is field attendance, physical visit evidence, meeting logs and manager oversight. The pilot calibrates thresholds and validates real-device reliability. Commercial sales modules are future work.

Phase	Deliverables and exit conditions	Indicative basis
0. Technical proof	Company-device matrix; GPS/camera tests; local storage/restart; sync UUIDs and server idempotency; country-scoped permissions. Gate: key risks validated.	2-3 weeks
1. Core MVP	Roles, device assignments, attendance/work calendars, organisations and branches, location/photos, visit/meeting logs, offline queue, GPS/device exceptions, approvals, basic dashboards, CSV.	4-6 weeks
2. Controls + pilot	Random/risk audit calls, anti-fraud flags, exception workbench, data-protection controls, import/export hardening and real-world one-market pilot. **No quote/order implementation.**	4-week pilot + fixes
3. Regional rollout	Tune market rules, staff training, support handoff, quarterly activity reports and remaining-country launch.	2-4 weeks
4. Future modules	Optional follow-up tasks, quotations, opportunity/order reporting, ERP/CRM integration and stronger device controls - separate approval/PRD.	Not committed

10.1 Pilot operating method

- Start with one market, representative company phones and a vetted organisation/branch list with approved pins.
- Weeks 1-2: measure GPS accuracy, sync failures, visit duration, exception reasons and baseline visits without automatic penalties.
- Weeks 3-4: evaluate team targets and flag handling; investigate false positives, approval capacity and offline failures.
- Before rollout, sign off privacy notices, training, manager procedures, backup recovery and device readiness.

10.2 Proposed pilot success measures

Measure	Operational acceptance threshold
Workdays	95%+ scheduled days have Start/End or approved leave/non-visit/exception; unresolved days surfaced.
Verified visit evidence	Publish verified share separately; 95%+ eligible attempts reach verified or reviewed exception state.
Meeting logs	95%+ submitted within 2 hours, apart from documented outages.
Offline syncing	95%+ same-day receipt where conditions permit; failed/lost events measured distinctly.
Client approvals	95%+ branch/pin requests closed within two working days.
Client audits	90%+ of conclusive sampled audit calls corroborate meetings; unreachable disclosed.
Exceptions	95%+ handled within configured SLA and decision audit complete.

> **Scope gate**
>
> No MVP acceptance gate depends on quotation accuracy, order verification, overdue follow-up completion, pipeline progression or revenue reporting.

11. Acceptance tests and field scenarios

These release-blocking scenarios must pass on approved company devices and, where applicable, in connected and offline modes.

ID	Test case and expected result
AT-01	Successful branch check-in with accurate GPS, live premises photo and required log yields exactly one Location Verified visit.
AT-02	Uncertain boundary reading triggers retry/exception, not automatic location verification.
AT-03	Check-out outside radius closes the visit and creates a visible flag.
AT-04	A second overlapping normal visit is blocked.
AT-05	Offline actions/media persist across normal app restart, then sync in dependency order.
AT-06	Repeated uploads of same event UUID produce only one server record.
AT-07	Offline clock anomalies are flagged without claiming trusted capture times.
AT-08	Lost browser/device data is not fabricated; salesperson can submit a retrospective unverified exception.
AT-09	Separate branches may share one parent company contact number without hard duplicate rejection.
AT-10	Rejected branch registration keeps history but does not become a verified visit.
AT-11	Manager-approved non-visit work counts toward attendance only.
AT-12	Only salesperson can submit missed/GPS-device exception logs; manager may approve/reject, not create.
AT-13	Client denial starts human dispute review, not automatic discipline.
AT-14	Cross-market direct API reads/writes are denied by server and RLS.
AT-15	MVP has no follow-up task, quotation, opportunity, reported-order creation, pricing or sales pipeline UI/API/KPIs.
AT-16	Delayed offline records reconcile after daily digest without losing earlier audit snapshots.

11.1 Required release evidence

Provide real-device QA evidence, security/RLS testing, successful backup restoration, offline failure/recovery logs, approved retention policy, support/escalation owner, pilot manager acceptance and a documented issue register.

12. Decisions, assumptions and open items

12.1 Agreed stakeholder decisions

Decision	Resolution for updated v1.2
Visits without client OTP	Count where location/photo/timing/log evidence passes. Independent client audit; denial triggers review.
GPS or phone failure	**Salesperson creates the log**, including after phone restoration; manager only approves/rejects. Approved location status remains unverified.
Company branches	Multiple distinct branches may share parent-company phone; duplicate checks warn without blanket blocking.
Non-visit workdays	Manager-approved legitimate office/virtual work counts for attendance, not physical visits.
Commercial features	**Excluded from MVP:** quotations, follow-up tasks/reminders, orders, opportunities, pipeline, conversion and revenue analytics. Meeting outcome/next-step notes only.
Document format	Retain Carbon Car Care visual style and preserve relevant operational detail.
Implementation strategy	Phased MVP, one-market pilot and separate future enhancements.
Device ownership	Company-issued phones with asset registration and replacement controls.
Integrations	Standalone MVP with versioned CSV; future ERP/CRM work requires separate approval.

12.2 Open decisions and responsible owners

Open decision	Proposed owner / before
Pilot market/team, approved client/branch master	Sales leadership / technical proof
Handset models, OS/browser matrix and asset/MDM policy	IT & Security / pilot readiness
Country calendars, holidays and legitimate non-field work	Country Sales Heads + HR / pilot
Employee notices, media retention, cross-border access	Legal / HR / IT Security / data capture
Photo-prohibited premises and evidence alternatives	Sales leadership + Legal / training
Geofence/accuracy thresholds, auditing, exception SLA	Pilot steering group / rollout
Support contacts, escalation, final timeline and resourcing	IT/Operations/Engineering / rollout
Any proposed future commercial feature set and ownership	Commercial/IT/Finance / separate future PRD

13. Future sales modules and ERP/CRM roadmap

The MVP is standalone. Neither commercial task/pipeline features nor live ERP/CRM connections are part of delivery, QA gates or pilot KPIs. The following is a possible later roadmap only.

13.1 Data foundations to retain at launch

- Use stable UUIDs for organisations, branches, client contacts, attendance, visits and meeting logs, with market, source and audit information.
- Publish versioned CSV exports with field types, country/timezone rules, event and branch IDs, and safe escaping.
- Retain evidence states and meeting narratives independently from any future opportunity, quotation or order system.
- Protect exports with role controls, download audit, limited retention and approved handling of client contact data.

13.2 Optional follow-on phases (not committed)

Future phase	Possible scope / entry condition
Discovery and ownership	Determine whether existing CRM/ERP should own quotations, follow-up reminders, orders and customer/account records; agree duplication policy.
Optional follow-up module	Only after new PRD: assign owner, due date, reminders, completion records, overdue queues and task KPIs.
Optional quotations and orders	Only after new PRD: formal request, value/currency, status, reference, manager verification and reconciliation.
Integration/API mapping	After field app stabilises: agree counterpart systems, country IDs, auth, ownership, updates and error/retry policies.
Pilot and reconciliation	Prove duplicate-safe exchange and matching to official systems of record; never treat salesperson notes as invoiced revenue.

13.3 Separate approval gates

- Commercial leadership, Finance, IT and Legal approve use cases, data responsibility, privacy and measurable success criteria.
- Engineering provides schema, UX design, permissions, QA/acceptance tests, capacity estimate and rollout plan.
- Do not create unused commercial tables or UI placeholders in MVP; keep only stable export contracts and extensible IDs.
- No automatic sales or customer merging until approved mapping, idempotency and rollback tests pass.

> **Current launch boundary**
>
> Version 1.2 MVP can ship and operate without commercial modules or ERP/CRM integration. Stable CSV exports are the only approved external hand-off.

Appendix A. Lovable MVP build specification

This appendix is the implementation prompt for the approved MVP. Build in staged, reviewable increments, validate SQL/RLS and Edge Functions, and do not infer future-phase sales functionality from meeting-log labels.

A1. Product and platform

> **Project brief**
>
> Build Carbon Field Sales, a mobile-first, offline-first standalone PWA for Carbon Car Care B2B salespeople on company-provided phones in Nigeria (Africa/Lagos), Zambia (Africa/Lusaka) and Malawi (Africa/Blantyre). Use Supabase Auth, Postgres, private Storage and Edge Functions. In MVP capture attendance, client/branch visits, evidence, meeting logs, exceptions, manager audit and CSV exports.

A2. Stage 1 - Roles, device management and clients

- Create roles `salesperson`, `sales_head`, `hq_admin`. Enforce country RLS and role checks on every privileged API route.
- Create company-owned device inventory with user assignment, approval/revocation and changed-device exception.
- Model organisations and their branches separately. Permit shared phone numbers across branches; show possible duplicates for review.
- Support validated bulk import, duplicate warnings, branch GPS pin proposals and country manager approval; log all changes.

A3. Stage 2 - Attendance and visit capture

- Salesperson Home actions: Start Day / Visit Client / Add Branch / Approved Non-Visit Day request / End Day / Sync and My Exceptions. Do **not** add Follow-ups/Quotes/Orders menus.
- Start Day captures company device session, GPS, live selfie and workday status. Country-specific calendars and approved absence apply.
- Visit action: select branch, fresh GPS + accuracy, geofence check, mandatory live premises photo, record check-in; retry or salesperson-created exception when GPS fails.
- Check-out always closes with GPS if available; if outside geofence, record location and flag. Only one normal visit can be open at once.
- Meeting log captures contact, purpose, products, narrative outcome, next-step text and notes. No quote/order value, due-date task or pipeline fields.
- Location Verified requires accepted GPS, live media, timing and completed log. Exception approval is Location Unverified.

A4. Stage 3 - Durable offline sync

- Cache the PWA shell plus minimum approved branch list. Queue media and events atomically in IndexedDB with unique UUID, session, sequence, timestamp and dependencies.
- Provide Offline / Pending / Syncing / Needs Review / Failed statuses, item-level retries and manual Sync Now.
- Edge Functions enforce idempotency, country/role permissions, geofence and event order, storing server receipt separately.
- No claim of tamper-proof browser data or trusted elapsed time across restarts; document unrecoverable storage loss.

Appendix A. Lovable MVP build (continued)

A5. Stage 4 - Logs, exceptions, audits and flags

- Meeting log must contain contact, purpose, products discussed, narrative outcome and next step; optional live-camera business-card photo and notes. No values, follow-up task records or commercial order states.
- Only authenticated **salesperson** may create and submit GPS/device-failure and missed-visit logs, including after access is restored. Country Sales Head may only review, approve/reject, comment or escalate.
- A manager-approved missing-GPS log remains **Manager Approved - Location Unverified**; preserve claimed time, submit time and reviewer trail.
- Implement flags for short visit, outside check-out, implausible travel, idle gap, late start/finish, no/low visit activity, late log, near start location, reused media, duplicates, clock/sync issues and unclosed day.
- Sample 5% of verified visits for weekly client audit plus risk-based prioritisation. Client Not Confirmed triggers a dispute, not automatic disciplinary action.

A6. Stage 5 - Dashboards and exports

- Daily manager view: scheduled/active/absent, last known sync, location-verified vs approved-unverified visits, branch approvals, missing logs and open flags.
- Weekly view: attendance, unique visited branches, verified activity, manager decisions, offline/sync health and audit/dispute outcomes.
- Quarterly view: organisation/branch coverage, verified-visit trends, repeat-visit distribution, flag and exception trends. No quote/order/pipeline charts.
- Provide filters by market/rep/period and audited CSV exports. Escape spreadsheet formula-leading values.
- Send 19:00 provisional daily exception digest and next-day corrected report, plus weekly HQ activity export.

A7. Stage 6 - Secure production readiness

- Private media and signed URLs; market RLS and API permissions; company handset inventory; immutable evidence and reviewer history.
- Approved retention/employee notice, cross-border privacy review, backup/restore test, monitoring and support escalation.
- Mobile sunlight-readable interface using Carbon blue #21409A, green #1B9A4A and yellow #E8CF2F; use supplied logo.
- Run AT-01 through AT-16 on approved real devices before pilot; document iOS/Android/browser compatibility.

> **Strict build exclusions**
>
> Do not generate MVP screens, database tables, Edge Functions, KPIs or automations for follow-up tasks, quotations, opportunities, reported orders, sales conversions, pipeline/revenue analytics or live ERP/CRM integration. These belong only to a separately commissioned future phase.

Appendix B. Glossary and version changes

B1. Operational definitions

Term	Meaning
Location Verified	Accepted GPS/geofence, live media, timestamp and meeting-log evidence; not automatically client-confirmed.
Client Confirmed	A later client audit supports that the claimed meeting occurred.
Approved Unverified Activity	Manager-approved legitimate visit/working activity where normal location evidence is missing.
Disputed Meeting	A client audit or review raises doubt, pending investigation and recorded resolution.
Provisional Offline	Saved on device but not yet received/validated by server; not final dashboard evidence.
Branch	Separate physical customer location under a parent organisation with its own pin and visit history.
Meeting Outcome	Descriptive field activity note; not a quotation, sales opportunity, order or financial status.
Free-text Next Step	Description written in the meeting log; does not create an assigned task or reminder.
Immutable Event	Original capture retained; later reviews and corrections are separate auditable records.
Sync Receipt	Server time a queued item arrives; may be later than actual capture.
RLS	Postgres Row Level Security; restricts access to records by role and market.

B2. Changes against original v1.1 and earlier v1.2

- Preserved the field activity purpose, three country markets, company handset decision and standalone/PWA architecture.
- Clarified conditional location verification, independent client audit, outside-geofence check-out and approved non-visit workdays.
- Introduced parent organisations and independently verified branches sharing company phone details.
- Confirmed that **only salespeople create** missed-visit and device/GPS exception logs; managers only approve/reject.
- **Moved all follow-up task, quotation, opportunity, order, pipeline and revenue reporting features out of MVP** into a future roadmap.
- Updated dashboard tables, data model, pilot phases, acceptance tests and Lovable build instructions to remove commercial launch dependencies.
- Replaced unrealistic offline tamper-proof assurances with event UUIDs, idempotent syncing, recovery and explicit data-loss limitations.

> **Document status**
>
> Carbon Field Sales Tracker PRD v1.2, MVP scope updated on 15 October 2026. Stakeholder sign-off remains subject to approved device models, pilot market, work calendars, retention policy and final delivery schedule.